Roman Brunetti

rbrunet@ncsu.edu • (828) 577-1838

EDUCATION

B.S. Electrical Engineering | North Carolina State University **Dec. 2027**

Relevant Coursework: Control Systems, Linear Systems, Electromechanical Energy Conversion, Engineering Dynamics, Microelectronics, Computer Systems Programming, Fundamentals of Logic Design, Calculus III

PROJECTS & RESEARCH

Helping Hand Project: Robotic Prosthetic Hand

- Electrical Hardware Lead within a university-funded, multidisciplinary organization
- Designing power electronics to support servo-stall current transients while maintaining logic signal integrity
- Using OrCAD X PSpice simulation to target voltage averages and ripples
- Reading datasheets and selecting electrical and mechanical components to meet design tolerances

Hand Rehabilitation Lab | NC State University **Sept. 2026-Present**

- Tested flexible EMG electrodes and a custom amplifier, checked PCB ground continuity, and recorded EMG signals; analyzed relaxed and contracted muscle activity in MATLAB using FFT and power spectra
- Wrote setup instructions for amplifier power, voltage checks, electrode placement, and signal recording

FPGA Reciprocal Cipher

- Built FPGA circuit to take a 4-bit input, a 4-bit key, and output a 4-bit cipher
- Designed and implemented two-way analog-to-digital converter logic
- Implemented several layers of toggleable reciprocal ciphers using Verilog language and Apio tool
- Used lab equipment to analyze and debug circuit

WORK EXPERIENCE

Orchard Inn Restaurant, Saluda NC | Fine Dining Server & Line Cook June **2022-Present**

- Built strong working relationships over 3+ years with the same team

SKILLS

Programming

- C, Verilog, MATLAB, LC-3 assembly language, Git and GitHub

Electrical

- Laplace techniques, impedance matching, transistor analysis, RLC circuits, three-phase, FPGA logic,
- PSpice (OrCAD X), basic PCB layout (Altium), breadboarding, through-hole soldering, multimeter, oscilloscope

Mechanical

- Rotating and translating dynamic system modeling, internal force analysis, SolidWorks, 3D printing, hand and power tools

Interests

Snowboarding | Mycology | Mountain biking | Tennis | Hiking | Camping | Piano | Cooking | Bookbinding | Reading | Woodworking